1.1.1. Regional Context and the Need for Early Warning

Flooding is a recurring hazard in Tunisia, affecting several regions under different rainfall and geographical conditions. Historical events include the September 2018 floods in Nabeul, severe rainfall episodes in the Sahel in September 2020, and earlier flooding affecting Greater Tunis and the Medjerda basin. These events highlight the importance of strengthening flood preparedness and early-warning capabilities across the country.

Within this broader context, the Cap Bon region, particularly the Nabeul Governorate, provides a relevant setting for examining the need for localized flood monitoring. The September 2018 Nabeul floods serve as a local historical reference for understanding the potential consequences of such events. Quantitative information on rainfall, damage, and human consequences should be supported by reliable historical sources.

1.1.2. Why Flash Floods Require a Specific Approach

Flash floods differ from slower-developing floods because water levels and flow conditions can change rapidly following intense rainfall. In small catchments, runoff can concentrate quickly in drainage channels, leaving limited time for people to react. Localized rainfall and the lack of continuous monitoring at some locations can also make it difficult to identify dangerous conditions in time.

Road crossings are particularly concerning because drivers and pedestrians may encounter rapidly rising water with little warning. These challenges emphasize the need for affordable, locally adaptable systems that connect hazard detection with timely warnings and appropriate action, including in areas with limited connectivity or infrastructure.

1.1.3. Illustrative Case Study: Oued Souhil, RR27 Crossing

Oued Souhil, near the RR27 crossing in the Nabeul region, is proposed as an illustrative site for exploring localized flash-flood monitoring. The described setting includes a sandy channel bed, nearby industrial buildings, and a road bridge used by vehicles. These features make the location relevant for investigating potential exposure and the feasibility of installing a monitoring system.

Figure 1. Proposed study area at Oued Souhil, near the RR27 crossing, Nabeul Governorate, Tunisia.

Methodological note: Oued Souhil is a preliminary candidate site. Its flood exposure and the feasibility of sensor deployment remain subject to field assessment.

1.1.4. Identified Community Need and Project Justification

The main community need is access to reliable, location-specific warnings early enough for people and local responders to act. A monitoring system should therefore do more than detect changes in water levels: it should communicate alerts to relevant recipients and support escalation when an initial warning receives no response.

The proposed project addresses this need through a low-cost, locally maintainable approach combining environmental monitoring, alert communication, and response confirmation. By connecting hazard detection with communication and follow-up action, the system aims to address the gap between identifying a potential flood threat and ensuring that the relevant people are informed and able to respond.

1.2 Existing Solutions and their gaps :

The Tunisia to Launch SMS Disaster Alert System as Part of GSMA 'We Care' :

Principle :
The project uses mobile technology to warn people about natural disasters before or while they occur, helping them react quickly and stay safe. It also supports the UN Sustainable Development Goals (SDGs), particularly those related to sustainable communities and disaster resilience.

Gap :

- The system cannot confirm whether alerts are received or understood, provide personalized guidance, or prevent false alarms from reducing public trust, and it does not ensure that people know how to respond effectively.

A LORAWAN-BASED IOT ARCHITECTURE FOR REAL-TIME FLOOD SENSING AND SAFETY NOTIFICATION VIA SMS :

Principle :
This study proposes a smart flood management system that automatically detects rising water levels and warns people when the risk of flooding reaches a certain threshold.

Gap :

- Limited disaster prediction: Monitoring current water levels can detect danger, but it may not predict a flood early enough for evacuation.

Strengthening hydromet and early warning systems and services in Tunisia:

Principle :
This roadmap assesses Tunisia's weather and hydrological services to identify gaps and improve early warning systems for floods, droughts, heatwaves, and other climate-related hazards.

Gap :

- The roadmap may not adequately address the timely dissemination of warnings and the provision of clear, actionable instructions to ensure that vulnerable populations can respond effectively to hazards.

Google Flood Hub: Principle and workflow

Principle:

Google Flood Hub uses artificial intelligence (AI), weather data, and hydrological models to predict river floods and identify areas at risk. Its goal is to provide early information so authorities and communities have time to prepare.

Gap :

- Important limitation: Flood Hub predicts risk; it does not guarantee that every person receives a warning. Forecasts are updated daily, and coverage and accuracy vary by location.

1.3 Measurable impact :

Metric	What it measures	Proposed Phase 2 Test Goals
Alert delivery rate	Percentage of alerts successfully delivered to intended recipients	At least 90%
Time-to-acknowledge	Time between alert delivery and confirmation from a recipient	Median under 1 minute
False-alarm rate	Percentage of generated alerts that are not associated with a genuine warning condition	Below 5%
Monitoring-node uptime	Percentage of the test period during which the monitoring device remains operational	At least 95%
Households covered	Number of households included in the pilot warning network	90 out of 100 households

1.4 Included SDGs :

- SDG 3 – Good Health and Well-Being: Early flood warnings help protect human lives, prevent injuries, and reduce health risks caused by flooding.

- SDG 6 – Clean Water and Sanitation: Flood alerts help authorities protect drinking-water sources and sanitation infrastructure from contamination and damage.

SDG 11 — Sustainable Cities and Communities

-The system helps protect residents, reduce flood damage, and improve community disaster preparedness.

- SDG 13 – Climate Action: The system strengthens preparedness and resilience against climate-related hazards, especially floods intensified by extreme rainfall.

1.5 Inclusion and Local Actors

Inclusive Warning Dissemination

An effective flood early-warning system must ensure that warnings reach all community members, regardless of age, income, digital literacy, or access to smartphones and internet connectivity. The proposed system adopts a multi-channel approach combining SMS notifications, local audible and visual alarms, and trained community volunteers.

At the monitored road crossing, a beacon and siren provide immediate local warnings, even when the gateway or communication network is unavailable. Registered volunteers receive alerts through SMS or handheld devices and can acknowledge them using a reply or a button. If an alert remains unacknowledged, the system automatically escalates it to the Municipality and then to Civil Protection. Volunteers can also help relay verified safety instructions to elderly residents and other people who may have difficulty receiving or understanding digital alerts.

Empowering Local Actors and Ensuring Sustainability

Local actors are essential to the reliability and long-term sustainability of the system. Municipal technicians can be trained to inspect the sensing nodes, check solar power and communication equipment, clean or replace components, and report faults. Clear maintenance procedures

and documentation can help local teams maintain the system without relying entirely on external specialists.

Community volunteers support warning dissemination and acknowledge alerts, while municipal officers and Civil Protection personnel remain responsible for their respective emergency-response duties. The system records alert delivery, acknowledgements, and escalation times, helping authorities evaluate how effectively the response chain operates. When a danger state is detected, it also generates a CAP 1.2 recommendation for the regional committee, which retains the authority to decide whether a public alert should be issued.

Local authorities, technicians, volunteers, and residents should be involved in selecting deployment locations, validating warning messages, and defining response procedures. This participatory approach aims to make the system accessible, locally maintainable , and better adapted to community needs .